

Miadul Islam Nizzan

Computer Science Undergraduate | AI/ML, NLP & AI-Driven Software
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Academic Profile

Computer Science and Engineering undergraduate with a strong academic record and project experience spanning artificial intelligence, machine learning, natural language processing, computer vision, multi-agent systems, software engineering, networking, cybersecurity, and embedded systems. Combines research-oriented development with student mentoring, IEEE leadership, technical documentation, and collaborative program delivery.

Education

Green University of Bangladesh	Expected January 2027
B.Sc. in Computer Science and Engineering	Rupganj, Narayanganj, Bangladesh
• Current CGPA: 3.70 / 4.00	
Royal Media College	2021
Higher Secondary Certificate (HSC)	Mymensingh, Bangladesh
• GPA: 4.89 / 5.00	
Shaheen Cadet School	2019
Secondary School Certificate (SSC)	Bhaluka, Mymensingh, Bangladesh
• GPA: 4.72 / 5.00	

Undergraduate Thesis (Ongoing)

Robustness, Explainability, and Attribution Stability of Lightweight Vision Transformers for Federated Chest X-ray Classification under Non-IID Client Distributions	2026 – Present
Green University of Bangladesh	GitHub

- Designing a federated binary-classification study for pleural effusion detection using NIH ChestX-ray14 across 10 simulated non-IID hospital clients.
- Comparing centralized, local-only, FedAvg, and FedProx training with DeiT-Tiny and ResNet-18, including noisy-label robustness and attribution-stability analysis.
- Applying Grad-CAM and Vision Transformer attribution methods to evaluate explanation consistency across client distributions.

Research Interests

Bangla natural language processing; social-media safety and online-harassment detection; retrieval-augmented generation; intelligent agents; computer vision; trustworthy and explainable AI; federated learning; AI for cybersecurity; network security.

Manuscript Submitted

Robustness and Tokenization Stability of Transformer Models for 12-Class Bangla Dialect Classification Under Character-Level Perturbations	Decision pending
Submitted to the 2026 IEEE 8th International Conference on Sustainable Technologies for Industry 5.0 (STI)	

- Evaluated three Transformer models on 59,453 BanglaDial records under progressive character-level perturbations.
- Found multilingual DistilBERT most robust at the 20% perturbation condition; typo-aware augmentation improved its accuracy from 51.005% to 62.913%.

Research in Progress

Multilevel Detection of Social Media Harassment Targeting Women	2026 – Present
Dataset development and text-classification model fine-tuning	
• Curating a multilevel dataset of social-media harassment targeting women and developing a structured labeling scheme for model training and evaluation.	
• Fine-tuning a text-classification model and preparing the study for a future research manuscript, subject to experimental results.	

Technical Expertise

AI, ML & NLP: TensorFlow, Keras, PyTorch, Hugging Face Transformers, scikit-learn, LangChain, RAG pipelines, multi-agent workflows, CNNs, MobileNetV2, BERT-based NLP, Bangla NLP
Agentic AI: LangChain, Groq API, Tavily search, tool calling, role-specific prompting, source attribution, Streamlit deployment
Federated & Explainable AI: FedAvg, FedProx, Non-IID learning, DeiT-Tiny, ResNet-18, Grad-CAM, Vision Transformer attribution
Programming: Python, JavaScript, TypeScript, Dart, PHP, SQL; working knowledge of C, C++, Java and x86 Assembly
Web & Mobile: React, Vite, Node.js, Django REST Framework, Flask, Flutter, REST APIs, HTML/CSS
Data & Databases: Pandas, NumPy, OpenCV, MySQL, PostgreSQL, SQLite, Supabase
Security & Networking: TCP/IP, packet analysis, AES, RSA, SHA-256, network fundamentals, cryptographic verification
Tools: Git, GitHub, Jupyter, Google Colab, VS Code, NetBeans, XAMPP, ~~TeX~~TeX, Microsoft Office

Selected Technical Projects

AI Research Assistant Agent

Python, LangChain, Groq, Tavily, Streamlit

[GitHub](#) | [Live Demo](#)

- Built a multi-agent research system in which a manager coordinates Researcher, Analyst, and Writer agents to produce structured, source-attributed research briefs.
- Added web-search tool calling, role-specific prompts, explicit failure handling, saved reports, and a Streamlit dashboard for execution tracking and research history.

CogniSearch: Research-Oriented AI Chatbot

Python, Streamlit, Groq API, Llama 3.3

[GitHub](#)

- Developed a modular conversational research workspace with specialized personas for AI/ML, cybersecurity, systems engineering, and general computer science.
- Implemented temperature, Top-P, and token controls, persistent conversation state, Markdown export, secure credential configuration, and separated UI and API/prompt modules.

Fosholer Bondhu: AI Agricultural Assistant

Python, TensorFlow, Keras, MobileNetV2, TensorFlow Lite

[GitHub](#)

- Developing a mobile AI assistant for crop-disease identification and treatment guidance for farmers in Bangladesh.
- Structured a MobileNetV2 transfer-learning workflow for disease classification, evaluation, and TensorFlow Lite deployment.

Smart Farming System: Bengali-First Agricultural Platform

React, TypeScript, Vite, Node.js, PostgreSQL/Supabase

[GitHub](#)

- Built a Bengali-first, local-first platform with a React/TypeScript frontend, Node.js API, local JSON persistence, file uploads, and optional PostgreSQL/Supabase connectivity.
- Added plant-disease detection, automated irrigation control, expert consultation workflows, and market-price verification.
- Integrated multi-provider AI chat, Bangla text-to-speech, and Server-Sent Events for real-time updates.

Farm Operation & Transaction Management

Flutter, Django REST Framework, JWT, SQLite, PostgreSQL

[GitHub](#)

- Built an offline-first dairy-farm system covering animal health, vaccination, milk production, inventory, transactions, loans, and reporting.

Secure File Sharing & Cryptographic Verification

Python, Flask, AES-128-CBC, RSA, SHA-256

[GitHub](#)

- Implemented AES-128-CBC, RSA, and SHA-256 from scratch in pure Python, plus role-based sharing, tamper simulation, audit logging, and automated tests.

Additional Projects

- **LangChain RAG Agent Prototype** – Built a data-connected Python agent and evaluated rapid LangChain development against lower-level API code. [LinkedIn](#)
- **AI Communication Systems** – Developed a web chatbot using Python sockets and an OpenAI API, plus a multi-user Java Swing messenger with SQLite and automated responses.
- **Operating System Simulation** – Implemented FCFS, SJF, and Round Robin scheduling, paging, deadlock detection, and performance analysis in C/C++.
- **Digital Logic & Embedded Systems** – Designed an ALU with a Python GUI and developed pulse-rate detection, fire suppression, and sensor-driven irrigation concepts.

Experience

Student Mentor

2025 – Present

Green University of Bangladesh

[Rupganj, Narayanganj, Bangladesh](#)

- Guide junior students in academic planning, technical coursework, projects, GitHub, presentations, and documentation.
- Support academic guidance sessions, workshops, and access to university resources.

Assistant Teacher

2023 – 2024

Kolatoli Darus Sunnah Al Arabiah Madrasah

[Rupganj, Dhaka, Bangladesh](#)

- Conducted structured lessons, mentored students, and developed instructional communication and classroom leadership skills.

Leadership & Professional Engagement

Vice Chair

2026 – Present

IEEE GUB Student Branch, Executive Committee 2026

- Coordinate technical events, workshops, student initiatives, communications, and cross-team collaboration.

Assistant Graphic Designer

2025 – 2026

IEEE GUB Student Branch, Executive Committee 2025

- Supported branding, event promotion, technical program publicity, and digital content development.

Community Leadership

2020

Dhumketu Blood Foundation; Dhitpur Boys Club

- Served as Joint Organizing Secretary and Event Organizing Secretary, coordinating outreach, events, and volunteers.

Awards, Membership & Academic Service

- **Dean's Award**, Green University of Bangladesh – outstanding academic performance.
- **Vice Chancellor's Award**, Green University of Bangladesh – academic excellence and consistent achievement.
- **Debate Champion**, Royal Media College – 2020 inter-college debate competition.
- IEEE Member, 2025 – Present.
- IEEE Day Ambassador – 2025 and 2026; volunteer at the STI 2025 Conference.

References

Mr. Syed Ahsanul Kabir

Chairperson & Associate Professor, Department of CSE, Green University of Bangladesh

kabir@cse.green.edu.bd

Professor Dr. Md. Ahsan Habib

Chairperson, Department of CSE, Bangladesh University of Business and Technology

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